Hyeryeon Park

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RESEARCH INTERESTS

- **Polymer nanocomposites and functional soft materials:** viscoelasticity, interfacial adhesion, and long-term reliability under repeated deformation
- **Particle assembly and patterning in polymer thin films:** spontaneous surface segregation, shear-flow-induced alignment, and magnetic-field-induced patterning
- **Processing–structure–property relationships in polymer films and coatings:** rheology, interfacial mechanics, and optical performance for flexible displays

EDUCATION

Mar 2025 – Present **Soongsil University** **Seoul, South Korea**

M.S. in Chemical Engineering, Expected Feb 2027

- GPA: 4.50/4.50
- Advisor: Jun Hyup Lee

Mar 2021 – Feb 2025 **Soongsil University** **Seoul, South Korea**

B.S. in Chemical Engineering

- GPA: 3.78/4.50
- Undergraduate Research Assistant, Nano/Micro Engineering Laboratory, Dec 2023 – Feb 2025

PUBLICATIONS

- **HR. Park**, JW. Kim and JH. Lee*, Development of Impact-Resistant Pressure-Sensitive Adhesive Based on Thermally Assisted Surface Enrichment of Modified Silica Nanoparticles. Manuscript in preparation.
- **HR. Park**†, JW. Kim†, JI. An†, WJ. Choi, YE. Yun and JH. Lee*, A Synergistic Dual-Network Strategy for Highly Recoverable and Optically Transparent Acrylic Adhesive Films. *ACS Applied Polymer Materials*, Under Review.
- JW. Kim†, **HR. Park**†, WJ. Choi, YE. Yun and JH. Lee*, Deciphering the long-term failure mechanism of the sealing framework in a proton exchange membrane fuel cell by assessing multi-stress degradation under accelerated simulated operating conditions. *Chemical Engineering Journal*, 2026, 538, 177005. 10.1016/j.cej.2026.177005
- JW. Kim†, DH. Kim†, **HR. Park**, GJ. Shin and JH. Lee*, Temperature-mediated controllable colorless polymer nanocomposite adhesives with remarkable optical transparency using sodium bicarbonate-embedded thermosensitive nanoparticles. *Chemical Engineering Journal*, 2025, 516, 164090. 10.1016/j.cej.2025.164090
- SY. Yang†, JW. Kim†, **HR. Park**, GJ. Shin and JH. Lee*, Highly stretchable and transparent hybrid double-network acrylic nanocomposite adhesive based on the synergy between stretchable silica nanoparticles and multifunctional dual crosslinkers. *European Polymer Journal*, 2025, 235, 114071. 10.1016/j.eurpolymj.2025.114071

† Equal contribution. * Corresponding author.

PATENT APPLICATIONS

- JH. Lee, JW. Kim and **HR. Park**, Korean Patent Application No. 10-2026-0040065, filed March 5, 2026.

GRANTS & FELLOWSHIPS

Sep 2025 – Aug 2026 **Master's Research Grant, National Research Foundation of Korea (NRF)**

- Selected through a competitive research proposal process for an independently proposed master's research project.

RESEARCH EXPERIENCE

Jun 2024 – Present

Development of Surface-Segregated Nanocomposite Adhesives and Overcoat Films for Impact Resistance

Sponsored by Nissan Chemical Korea

- Developed surface-functionalized nanoparticle/PSA systems with controlled surface segregation, achieving a 20% improvement in impact resistance and leading to a Korean patent application.
- Extended the surface-segregation strategy to overcoat films with a hard-surface/soft-bulk mechanical gradient by controlling nanoparticle enrichment.
- Quantified nanoparticle distribution and correlated segregation behavior with mechanical and optical performance using SEM/EDS, nanoindentation, ball-drop testing, and UV–Vis spectroscopy.

Jul 2026 – Present

Development of High-Impact-Resistant Pressure-Sensitive Adhesives via Shear Flow-Induced Nano-Alignment of Nanorods

Sponsored by Samsung Display Corporation

- Designed and conducted systematic studies of anisotropic nanoparticle morphology to elucidate shear-induced nanorod alignment in polymer resin formulations.
- Optimized nanoparticle loading, resin viscosity, and blade/roll-to-roll coating parameters to balance optical transparency, haze, adhesion, and impact resistance.

Jul 2024 – Present

Development of Highly Durable Adhesive for Fuel Cell MEA and Reliability Evaluation Method

Sponsored by Hyundai Motor Group

- Designed and executed accelerated multi-stress degradation protocols to evaluate the long-term durability of polymeric sealing materials for fuel-cell MEAs.
- Formulated and optimized adhesive resin systems and their rheological properties for jetting, screen printing, and spray-coating processes.

May 2026 – Aug 2026

Development of Highly Recoverable Dual-Network Acrylic Adhesive Films for Flexible Displays

- Developed dual-network acrylic adhesive films using elastic nano-crosslinkers and molecular-bridge crosslinking to improve mechanical recovery while maintaining optical transparency and adhesion.
- Designed UTM-based stress-relaxation, strain-recovery, and cyclic loading–unloading tests across multiple strain and cycle conditions to evaluate viscoelastic response and cyclic mechanical stability.

Sep 2025 – Aug 2026

Development of Stretchable and Transparent Optical Films with Biomimetic Surface Patterns for Free-form Displays

Supported by the NRF Master's Research Grant

- Designed and synthesized highly stretchable, transparent polyurethane acrylate (PUA) resin systems for bioinspired free-form display applications.
- Investigated an inkjet-based micropatterning process for stretchable optical films, targeting an elongation of >600%, immediate strain recovery of >90%, and optical transmittance of >90%.

Mar 2026 – Present

High-Resolution Patterning of Ferromagnetic Particles in Organic Thin Films Using a Tunable Magnetic Polarity Array Based on the Magnetic Trapping Effect

Sponsored by the Ministry of Science and ICT (MSIT) of Korea

- Co-authored the research proposal and designed the particle surface-modification strategy, including particle selection and functional-group design, to enable magnetic trapping and patterning in organic thin films.

TEACHING EXPERIENCE

Sep 2026 – Present **SEM/EDS Instrumentation Assistant**

Soongsil University, Research Equipment Center **Seoul, South Korea**

- Operated and maintained FE-SEM/EDS instrumentation and assisted users with instrument operation, troubleshooting, and routine technical issues.
- Trained new users in sample preparation, instrument operation, and basic SEM/EDS data acquisition.

Mar 2026 – Aug 2026 **Lead Teaching Assistant**

Soongsil University, Department of Chemical Engineering **Seoul, South Korea**

- Supervised and coordinated a team of teaching assistants for Basic Chemical Engineering Laboratory and Chemical Engineering Laboratory courses serving approximately 220 students.
- Delivered laboratory lectures and coordinated TA scheduling, grading, and experiment preparation across both courses.
- Served as the department-level safety coordinator for undergraduate chemical engineering laboratories, overseeing laboratory safety procedures and compliance.

Mar 2025 – Feb 2026 **Teaching Assistant**

Soongsil University, Department of Chemical Engineering **Seoul, South Korea**

- Assisted with Basic Chemical Engineering Laboratory courses for second-year Chemical Engineering students.
- Delivered lectures and led grading and review sessions for 108 students.

SKILLS & TECHNIQUES

- **Materials Synthesis & Formulation:** Polymer resin formulation, nanoparticle surface functionalization, polyurethane acrylate synthesis, nanocomposite formulation
- **Coating & Patterning:** Blade coating, roll-to-roll coating, spin coating, jetting, screen printing, spray coating
- **Characterization:** FE-SEM/EDS, AFM, FT-IR, UV–Vis, TGA, DSC, XRD, DLS, nanoindentation, UTM, optical microscopy (OM), contact angle
- **Software & Data Analysis:** OriginPro, SigmaPlot, ImageJ, Rhino; rheological analysis, particle-distribution analysis, structure–property relationship analysis

ADDITIONAL EXPERIENCE

Dec 2019 – Jul 2023 **Chemistry Assessment Development Team Lead**

VERADI Co., Ltd. **Seoul, South Korea**

- Developed and reviewed high school chemistry practice assessment items for Korea's national college entrance examination (CSAT), focusing on scientific accuracy, clarity, and difficulty calibration.
- Led the chemistry item-development team, coordinating question writing, peer review, revision, and quality control for published chemistry mock-exam materials, including the *Helios* series.

VOLUNTEER EXPERIENCE

Sep 2023 – Apr 2024 **Volunteer**

Beautiful Store, Songpa Garak Branch **Seoul, South Korea**

- Supported store operations including sales, customer assistance, and store organization.
- Received, sorted, and organized donated goods and prepared items for transfer to the organization's distribution network.